

* Bluetooth.

- It is a short-range wireless communication technology used for data exchange between devices using radio waves in the 2.4 GHz ISM band.

* Bluetooth Master.

- The device that initiates and controls the bluetooth connection.

* Bluetooth slave.

- Devices that respond to the master and perform specific actions.

* Piconet

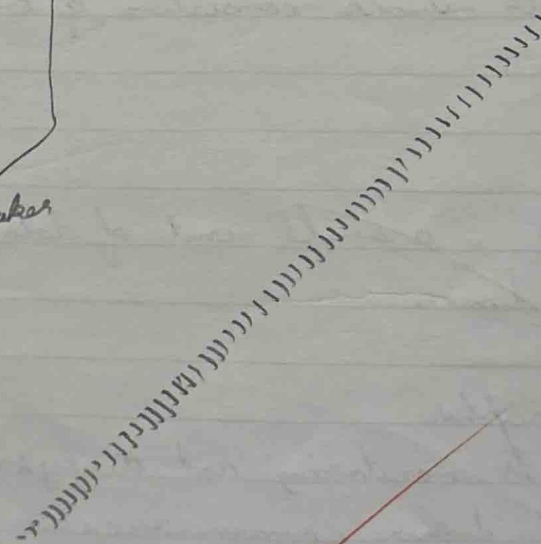
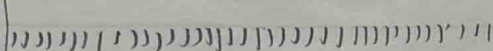
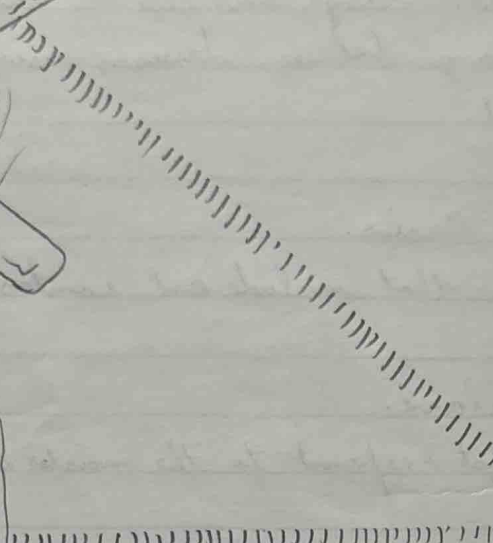
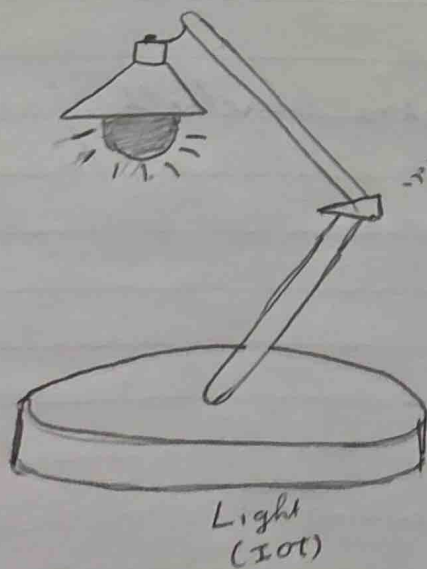
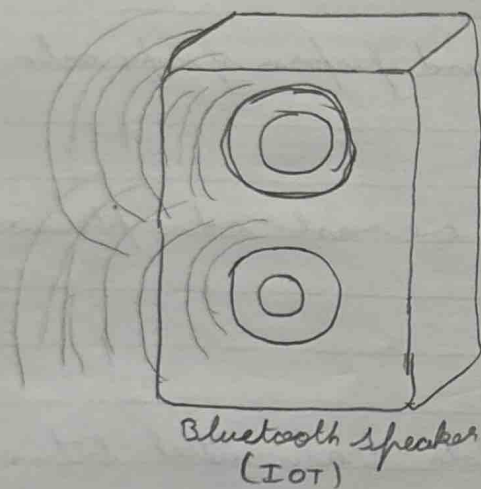
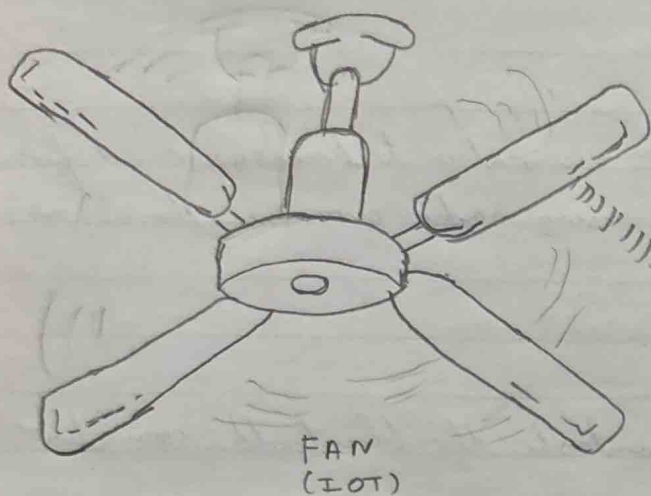
- A Bluetooth network consisting of one master and up to seven slave devices.

* Packet :

- A packet is a small unit of data transmitted between bluetooth devices.

* Packet Capture

- It refers to monitoring and analysing bluetooth packets exchanged during communication.



* Advantages

- Low power consumption
- No need for internet connectivity
- Simple device pairing
- Low cost implementation

* Disadvantages

- Limited communication range.
- Lower data rate compared to wi-fi.
- Prone to interference in 2.4 GHz band.
- Supports limited number of devices.

* Applications

- Wireless audio devices
- Smart home devices
- Wearable devices
- IOT and automation systems.

* Devices used:

- Portable Music player
- Bluetooth speaker
- Fan, light
- Cisco packet tracker.

* Procedure.

- Drag and place the portable Music player device.
- Add Bluetooth-enabled IOT devices
- Ensure all devices are within Bluetooth communication range

- Pair the bluetooth devices with the portable music player.
- switch to simulation mode.
- Send control commands from the master device.
- Observe and verify device response to commands.

* Observation

- Bluetooth devices were successfully paired.
- Devices responded correctly in bluetooth commands.
- Packet movement was visible in simulation mode.

* Result :

- The study & demonstration of bluetooth communication were successfully carried out using CPT.
- Packet capture confirmed reliable data transmission b/w bluetooth-enabled IOT devices.

* Conclusion

- This experiment demonstrates the working of Bluetooth communication where a master device controls multiple slave devices within a short range. Bluetooth provides a low-powered and cost-effective solution for personal area networks and IOT applications.

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